



Recent and chronic ambient air pollution exposure in association with telomere length and mitochondrial DNA content in the general population

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ABSTRACT

Telomere length (TL) and mitochondrial DNA content (mtDNAC) are biomarkers of biological ageing that respond to multiple stressors, including air pollution. Despite growing research interest, the association between recent and chronic air pollution and these biomarkers in the general population remains unclear. This study investigated the association between air pollution exposure and TL and mtDNAC using data from the 2018 Belgian Health Examination Survey. Multivariable adjusted generalised linear mixed models were applied to assess the exposure to nitrogen dioxide (NO₂), fine particulate matter ≤ 2.5 μm (PM_{2.5}), and black carbon (BC) over 1-week (recent) and 1-year (chronic) periods prior to participation, estimated with a high-resolution spatiotemporal model. Leucocyte TL and mtDNAC were measured using qPCR. A total of 756 participants (mean age 50.6 years, 49.9 % women) were included in the study. Recent exposure to PM_{2.5} was associated with a 2.40 % (95 % CI: 0.16, 4.69; $p = 0.036$) longer TL per IQR increment. Trends of lower mtDNAC were observed for chronic exposure to BC (−3.11, 95 % CI: −6.19, 0.07; $p = 0.055$) and NO₂ (−4.02, 95 % CI: −8.22, 0.36; $p = 0.072$) per IQR increment. No significant associations were observed between chronic air pollution and TL or recent exposure and mtDNAC. These results suggest an inverse association between chronic air pollution and mtDNAC, and a positive association between recent exposure and TL, providing insight into the time-sensitive and air pollutant effects on ageing biomarkers.

1. Introduction

Ambient particulate matter (PM) pollution exposure is recognised as the leading environmental risk factor affecting life expectancy, as reported in the latest Global Burden of Disease (GBD) report (Sang et al., 2022). Gaining a better understanding of biological targets for air pollution may provide insights into how air pollution leads to adverse health effects. Biological ageing is a key risk factor for common diseases like cancer, dementia and cardiovascular conditions (Niccoli and Partridge, 2012). Although substantial research has been conducted over the past decade, evidence continues to grow, suggesting that the hallmarks of biological ageing could help explain the link between air

pollution and adverse health outcomes (Jylhava et al., 2017; Peters et al., 2021).

Two biomarkers of ageing commonly used in epidemiological research are telomere length (TL) and mitochondrial DNA content (mtDNAC). Telomeres, which are the protective caps at the ends of chromosomes, consist of repeated TTAGGG sequences which shorten with each cell division due to the ‘end-replication problem’ (Wellinger, 2014). Mitochondria, often referred to as the cell’s powerhouses, are essential for energy production, apoptosis, innate immune response and cell signalling (Shaughnessy et al., 2014). The close relationship between telomeres and mitochondrial function has led to the formulation of the telomere-mitochondrial ageing hypothesis (Sahin et al., 2011).

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Observational studies linking ambient air pollution with TL and mtDNAc suggest that these biomarkers may offer a possible biological mechanism by which environmental exposures contribute to ageing, health and disease (Martens and Nawrot, 2016; Van Der Stukken et al., 2023).

Previous research on adults has revealed associations between air pollution exposure and differences in TL and mtDNAc. However, findings regarding the direction of the association for different time windows and air pollutants are inconsistent (Mishra et al., 2024). For instance, short-term exposure (up to a week) has been associated with a longer TL for PM_{2.5} (Hou et al., 2012; Pieters et al., 2016) and a higher mtDNAc for BC (Zhong et al., 2016) and polycyclic aromatic hydrocarbons (Pavanello et al., 2013). In contrast, short-term exposure to NO₂ (Everson et al., 2020) and benzene (Hoxha et al., 2009) has been associated with shorter TL, while a lower mtDNAc was found with NO₂ exposure (Bai et al., 2018). Regarding chronic exposure, PM_{2.5} and BC have been associated with shorter TL in men (Pieters et al., 2015; Ward-Caviness et al., 2016) and reduced mtDNAc with PM_{2.5} (Pieters et al., 2015), while indoor coal smoke exposure has been associated with longer TL (Xue et al., 2020). However, a large-scale analysis in a contemporary cohort from the UK Biobank found no significant association between various air pollutants and TL (Bountziouka et al., 2023).

In this study, we examined the association between recent and chronic exposure to multiple air pollutants (fine particulate matter ≤ 2.5 μm (PM_{2.5}), BC and NO₂) and the ageing biomarkers TL and mtDNAc in adults.

2. Materials and methods

2.1. Study population

This study was conducted as part of the Belgian Health Examination Survey (BELHES), a subset ($n = 1184$) of the 2018 Belgian Health Interview Survey (BHIS) (Hautekiet et al., 2020). Participants were recruited using a multistage sampling design, with the Belgian National Registry serving as the sampling frame. Eligible BHIS participants were invited to join the BELHES until regional quotas were met. During the study, participants underwent a health examination that included anthropological measurements, and trained nurses collected blood and urine samples (Nguyen et al., 2020).

For the present study, 756 BELHES participants were included. Individuals were excluded if they had recently moved to their current address (in 2017 or 2018) ($n = 140$) to ensure accurate chronic air pollution exposure data. Participants were also excluded if either TL or mtDNAc could not be measured ($n = 205$). Specific reasons for exclusion were: no blood sample provided ($n = 104$), no consent for DNA analysis ($n = 33$), blood sample quality issues (low volume ($n = 10$), contamination ($n = 1$), broken tubes ($n = 17$)), or high technical variability in quantitative real-time PCR (qPCR) replicates ($n = 40$). Additionally, 83 participants lacked data on key covariates and were excluded from further analysis.

As part of the BELHES, this project was approved by the Medical Ethics Committee of the University Hospital Ghent (registration number B670201834895) and conducted in accordance with the recommendations of the Belgian Privacy Commission. All participants provided written informed consent, and the study adhered to the Helsinki Declaration.

2.2. Residential air pollution exposure

To estimate daily exposure to PM_{2.5}, black carbon (BC), and nitrogen dioxide (NO₂) at the participants' residences, we used a high-resolution spatiotemporal interpolation model (Janssen et al., 2008). This model combined air pollution data from fixed monitoring stations in Belgium with CORINE Land Cover (CLC) information and a dispersion model that included both point and line pollution sources (Lefebvre et al., 2013; Lefebvre et al., 2011; Maiheu B et al., 2013). The model's performance

was validated using leave-one-out cross-validation, with 34 monitoring points for PM_{2.5}, 44 for NO₂, and 14 for BC. The model was found to explain 80 % of the variability for PM_{2.5}, 78 % for NO₂, and 74 % for BC (Maiheu B et al., 2013; Lefebvre et al., 2011).

Previous studies have shown positive correlations between modelled chronic residential BC exposure and internal BC loads in urine (partial $r = 0.17$) (Saenen et al., 2017), placental tissue (partial $r = 0.50$), maternal blood (partial $r = 0.57$), and cord blood (partial $r = 0.68$) (Bongaerts et al., 2022). This study used the daily exposure values to calculate mean residential exposure during the week or year before participation.

2.3. Telomere length and mitochondrial DNA content assay

Blood samples were centrifuged for 15 min at 3000 rpm and stored at -80 °C. DNA was extracted from the buffy coat using the QIAgen Mini Kit (Qiagen, N.V.V Venlo, The Netherlands), with purity and quantity measured by a NanoDrop spectrophotometer (ND-2000; Thermo Fisher Scientific, Wilmington, DE, U.S.A.), and DNA integrity checked via agarose gel electrophoresis. Samples were standardised to a DNA input of 6 ng for each qPCR reaction using the Quant-iT™ PicoGreen® dsDNA Assay Kit (Life Technologies, Europe).

Relative TL and mtDNAc were measured in triplicate using a qPCR assay, as described previously (Hautekiet et al., 2022). Briefly, reactions were conducted in a 384-well format on a 7900HT Fast Real-Time PCR System (Applied Biosystems, Foster City, CA, USA) using telomere, mtDNAc and single copy-gene reaction mixtures and PCR cycles as described in [supplementary text 1](#). Different quality control measures were taken into account, including reaction efficiency (90–110 %), inter-run variability ($n = 6$), and non-template controls ($n = 3$), and data was normalized to the single-copy gene using qBasePLUS software (Biogazelle, Zwijnaarde, Belgium).

Leucocyte TL and mtDNAc were expressed as the ratio of telomere copy number to the single-copy gene number (T/S) and mtDNA copy number to the single-copy gene number (M/S), respectively, relative to the population means. Assay reliability was confirmed by interclass correlation coefficients (ICCs) of the technical replicates, inter-run calibrators as proposed by the Telomere Research Network, using RStudio version 1.1.463 (RStudio PBC, Boston, MA, United States). The intra-plate ICCs of T/S ratios, TL runs, M/S ratios, mtDNAc runs and single-copy runs were respectively 0.80 ($p < 0.0001$), 0.91 ($p < 0.0001$), 0.82 ($p < 0.0001$), 0.92 ($p < 0.0001$) and 0.78 ($p < 0.0001$).

2.4. Statistical analysis

Statistics were conducted using SAS software (version 9.4; SAS Institute Inc., Cary, NC, USA). TL and mtDNAc were $\log(10)$ transformed to reduce skewness and achieve a more normal distribution. We evaluated recent (exposure during the week before examination) and chronic (exposure during the year before examination) exposure to three types of air pollution (PM_{2.5}, BC and NO₂). Linear mixed models (GLIMMIX; unstructured covariance matrix) were used for each air pollutant separately but included both exposure time windows. The results are reported as the percentage change in TL or mtDNAc (95 % CI) per IQR increment in air pollution.

A priori selected covariates included as fixed effects were age (continuous), sex (male; female), highest educational level of the household (up to lower secondary; higher secondary; college or university), country of birth (Belgium; EU; non-EU), region (Flanders; Brussels Capital Region; Wallonia), season (spring; summer; autumn, winter), smoking status (current smoker; current non-smoker), BMI (continuous) and run plate number (1–12). Since multiple members from the same household participated in the BELHES, we included household number as a random effect in the model.

Bivariate analyses of the associations between covariates and TL or mtDNAc were conducted using the same model. Unadjusted Spearman's

correlations assessed the associations between the air pollutants and different exposure windows. Chi-squared tests (for categorical data) and t-tests (for continuous data) were used to compare the characteristics of included and excluded participants.

2.5. Sensitivity analysis

In a first sensitivity analysis, we explored whether physical activity (intensive training/sport ≥ 4 h per week; light activities/sport < 4 h; sedentary activities), having a chronic disease (yes; no), hypertension, measured as systolic blood pressure ≥ 140 or diastolic blood pressure ≥ 90 (yes; no), or percentage green space in a 1000 m radius around the residence (continuous) may change the association between air pollution and the biomarkers of ageing. Green space was calculated with CORINE Land Cover data of 2012 (European Environment Agency) and included the classes of forest and semi-natural areas and artificial, non-agricultural vegetated areas. These three parameters were added separately to the main model as fixed covariates.

In a second sensitivity analysis, we evaluated the difference between subjective and objective measures by replacing smoking status with urinary cotinine levels as a covariate. Determination of urinary cotinine was performed by online solid phase extraction-ultra performance liquid chromatography-tandem mass spectrometry (SPE-UPLC MS/MS) as described by (De Cremer et al., 2013). Cotinine was adjusted for creatinine ($\mu\text{g/g}$). Creatinine concentrations were analysed using a kinetic colorimetric assay based on the Jaffé method (reagent: Creatinine Jaffé Gen.2 ref. 064071370190, Roche Diagnostics GmbH, Mannheim, Germany). Participants with a creatinine level below 0.3 g/l and above 3 g/l were excluded (World Health Organization, 1996). A cotinine cut-off of 100 $\mu\text{g/g}$ creatinine was applied to distinguish current smokers from current non-smokers (Paci et al., 2018).

3. Results

3.1. Population characteristics

Table 1 summarizes the characteristics of the study population. Women comprised 49 % of the participants, with a mean age (SD) of 50.5 (15.5) years. Most participants were born in Belgium (82.3 %), and 58.1 % had a college or university degree as the highest educational level in the household. Regional representation included 49.7 % from Flanders, 20.4 % from Brussels Capital Region, and 29.9 % from Wallonia. Participants were enrolled in different seasons: 17.7 % in spring, 25.0 % in summer, 49.5 % in autumn and 7.8 % in winter. Current smokers comprised 20.5 % of the study population, and the mean BMI (SD) was 26.3 (4.7) kg/m^2 . The mean (SD) relative TL was 1.02 (0.22), and mtDNAc was 1.03 (0.24), with a positive correlation between the ageing biomarkers (Spearman's correlation = 0.17, $p < 0.0001$).

A comparison of the characteristics between the included and excluded participants is presented in Supplementary Table 1. In the included versus the excluded participants, we found more women (respectively 49.9 % vs 57.7 %, $p = 0.0094$), and higher educated households (58.1 % vs 47.3 %, $p = 0.0019$), a more significant number of participants living in Flanders (49.7 % vs 39.7 %, $p = 0.0011$) and born in Belgium (82.3 % vs 76.1 %, $p = 0.00017$), and more participants with a lower BMI (mean 26.3 vs 27.1, $p = 0.014$). No differences were found in age, season, or smoking status.

Supplementary Table 2 shows the bivariate associations between demographic characteristics and both biomarkers. In summary, men had a 5.79 % (95 % CI: 8.44 to -3.06 , $p < 0.001$) shorter TL and an 8.63 % (95 % CI: 11.47 to -5.70 , $p < 0.0001$) decrease in mtDNAc compared to women. Each additional year of age was associated with a 0.59 % (-0.68 to -0.50 %, $p < 0.0001$) decrease in TL, and a 0.12 % (95 % CI: 0.23 to -0.01 %, $p = 0.026$) lower mtDNAc. Individuals from households with a college or university education as the highest level attained had an 11.65 % (95 % CI: 6.51 to 17.04, $p < 0.0001$) longer TL and a

Table 1

Characteristics of the study population (n = 756).

Characteristics	n (%) or mean (SD)
Women	377 (49.9 %)
Age (years)	50.6 (15.6)
Highest educational level in the household	
Up to lower secondary school	103 (13.6 %)
Higher secondary school	214 (28.3 %)
College or university	439 (58.1 %)
Country of birth	
Belgium	622 (82.3 %)
EU	74 (9.8 %)
Non-EU	60 (7.9 %)
Region	
Flanders	376 (49.7 %)
Brussels Capital Region	154 (20.4 %)
Wallonia	226 (29.9 %)
Season	
Spring	134 (17.7 %)
Summer	189 (25.0 %)
Autumn	374 (49.5 %)
Winter	59 (7.8 %)
Current smoker	155 (20.5 %)
BMI (kg/m^2)	26.3 (4.7)
Physical activity^a	
Intensive training/sport ≥ 4 h per week	157 (22.0 %)
Light activities/sport < 4 h	373 (52.2 %)
Sedentary activities	184 (25.8 %)
Green space	6.7 (10.3)
Cotinine > 100 $\mu\text{g/g}$ creatinine^b	141 (20.0 %)
Chronic disease^c	214 (28.0 %)

^a n = 714.

^b n = 706.

^c n = 743.

9.50 % (95 % CI: 4.09 to 15.19, $p = 0.00052$) higher mtDNAc compared to those in the lower educational category. Current smokers had a 8.33 % (95 % CI: 12.01 to -4.49 , $p < 0.0001$) lower mtDNAc, though no significant association was found with TL. These findings were further supported by bivariate analysis between urinary cotinine levels and both mtDNA content and TL. Additionally, no significant association was observed between the country of birth or region and the biomarkers.

3.2. Exposure characteristics

The mean (interquartile range, IQR) for recent exposure levels of $\text{PM}_{2.5}$, BC and NO_2 were 12.1 (7.4) $\mu\text{g/m}^3$, 1.0 (0.6) $\mu\text{g/m}^3$ and 18.9 (10.1) $\mu\text{g/m}^3$, respectively (Table 2). For chronic exposure, the mean (IQR) was 12.0 (3.0) $\mu\text{g/m}^3$ for $\text{PM}_{2.5}$, 0.9 (0.3) $\mu\text{g/m}^3$ for BC and 19.0 (8.1) $\mu\text{g/m}^3$ for NO_2 . Positive correlations were observed between the air pollutants across different time windows (Suppl. Table 3), with the strongest correlation observed between chronic BC and NO_2 levels (Spearman's correlation = 0.96, $p < 0.0001$) (Suppl. Table 3).

3.3. Air pollution and the biomarkers of ageing

The results for the association of recent and chronic air pollution

Table 2

Ambient residential exposure characteristics ($\mu\text{g/m}^3$) (n = 756).

		Mean	25th percentile	Median	75th percentile	IQR
$\text{PM}_{2.5}$	Recent	12.1	7.9	10.7	15.3	7.4
	Chronic	12.0	10.6	12.8	13.6	3.0
BC	Recent	1.0	0.7	0.9	1.3	0.6
	Chronic	0.9	0.8	0.9	1.1	0.3
NO_2	Recent	18.9	13.4	18.1	23.5	10.1
	Chronic	19.0	15.1	17.8	23.2	8.1

$\text{PM}_{2.5}$ = fine particulate matter ≤ 2.5 μm , BC = Black carbon, NO_2 = nitrogen dioxide. IQR = interquartile range.

with both TL and mtDNA are shown in Figs. 1 and 2, respectively, and estimates for both unadjusted and fully adjusted models are presented in Table 3. For the fully adjusted models, an IQR increment ($7.4 \mu\text{g}/\text{m}^3$) in recent $\text{PM}_{2.5}$ was associated with a 2.40 % (95 % CI: 0.16, 4.69; $p = 0.036$) longer TL, but no association was found for chronic exposure or the other air pollutants. On the other hand, an IQR increment in chronic BC ($0.3 \mu\text{g}/\text{m}^3$) and NO_2 ($8.1 \mu\text{g}/\text{m}^3$) was associated with a respectively -3.11 % (95 % CI: 6.19, 0.07; $p = 0.055$) and -4.02 % (95 % CI: 8.22, 0.36; $p = 0.072$) lower mtDNA. No association was found for recent exposure to BC, NO_2 or exposure to $\text{PM}_{2.5}$.

3.4. Sensitivity analysis

We determined the impact of physical activity, chronic disease, and green space on the association between air pollution and the biomarkers. The results of these analyses did not show strong confounding or substantial changes to the main model's estimates (Suppl. Table 4). Adjusting for cotinine instead of smoking status resulted in slightly stronger estimates for TL, but no other differences were found (Suppl. Table 4).

4. Discussion

In this study, chronic ambient residential exposure to BC and NO_2 showed a trend towards an association with a lower mtDNA. No association with mtDNA was found for recent exposure. Second, chronic exposure to ambient air pollution was not associated with TL, but recent exposure to $\text{PM}_{2.5}$ was associated with a longer leucocyte TL.

4.1. Mitochondrial DNA content

The present study found lower mtDNA levels associated with higher chronic exposure to air pollution, consistent with previous findings. Similar inverse associations were reported for $\text{PM}_{2.5}$, PM_{10} and NO_2 in Chinese adults (Li et al., 2021), for $\text{PM}_{2.5}$ and PM_{10} in women (Wang et al., 2020) and for a multiple exposure model (including recent, sub-chronic and chronic exposure) for $\text{PM}_{2.5}$ in elderly (Pieters et al., 2015). Our previous study with children aged 9–12 also showed that chronic $\text{PM}_{2.5}$ exposure was associated with lower mtDNA (Hautekiet et al., 2021). However, similar to the present study in adults, we did not observe a clear association between recent air pollution exposure and mtDNA. Studies on recent exposures have yielded mixed results. For example, Xia and colleagues found no association between mtDNA and $\text{PM}_{2.5}$, PM_{10} or NO_2 in diabetes patients (Xia et al., 2015), while a study

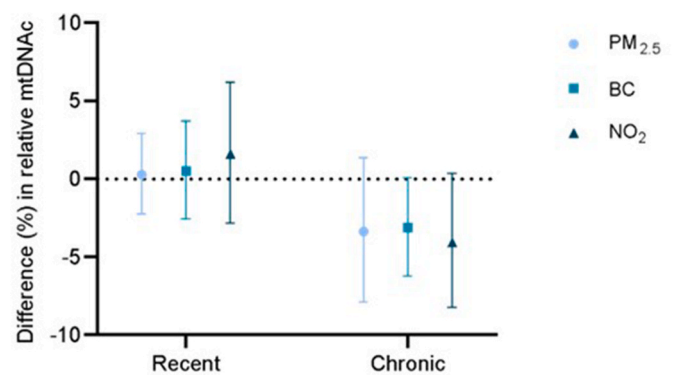


Fig. 2. Difference (%) with 95 %CI in average relative mitochondrial DNA content (mtDNA) (with 95 % CI) in association with recent (week) and chronic (year) exposure to $\text{PM}_{2.5}$, BC and NO_2 . Based on multiple exposure models where the exposure windows were fitted in the same model. Estimates were adjusted for age, sex, highest educational level of the household, country of birth, region, season, smoking status, BMI and plate number as fixed covariates and for household number in the random statement.

of Beijing office workers and truck drivers reported a negative association between mtDNA recent elemental carbon and PM_{10} but not with $\text{PM}_{2.5}$ exposure (Hou et al., 2013). Other research has found lower mtDNA levels with exposure to $\text{PM}_{2.5}$ in healthy non-smoking women (Wong et al., 2017) and NO_2 in older people (Bai et al., 2018). Conversely, positive associations between mtDNA and air pollution were found for older men and male steel workers exposed to ambient BC levels (Zhong et al., 2016) and particulate matter (Hou et al., 2010).

Our findings show an inverse association between chronic air pollution exposure and mtDNA but no significant link with recent exposure. This difference may arise from distinct biological mechanisms operating over different time windows. Inflammation and oxidative stress are critical processes connecting air pollution to human health, as noted in several reviews (Fiordelisi et al., 2017; Moller et al., 2014; Valavanidis et al., 2013). Air pollution has been associated with mitochondrial DNA damage through oxidative stress (Grevendonk et al., 2016). Higher levels of reactive oxygen species (ROS) can cause and result from mitochondrial damage and dysfunction (Martens and Nawrot, 2016). Mitochondrial DNA is especially vulnerable to damage due to its less efficient repair mechanisms than nuclear DNA (Liu and Demple, 2010), making it a primary target of ROS. Changes in mtDNA may reflect the body's response to external stressors or triggers (Hou et al., 2010). Mitochondria may initially increase in number and mtDNA to compensate for short-term damage and energy deficiency (Hou et al., 2010). Over time, however, the bioenergetic and replicative capacity of damaged mitochondria decreases, leading to reduced mtDNA, loss of mitochondrial function, cell senescence and ultimately, apoptosis (Hou et al., 2010; Li et al., 2013; Wong et al., 2009). These findings suggest that mtDNA may reflect the cumulative oxidative stress burden of chronic air pollution exposure.

4.2. Telomere length

In line with our research, Pieters et al. (2015) evaluated chronic $\text{PM}_{2.5}$ exposure in older people in association with TL. Their multiple exposure model (including recent, sub-chronic and chronic exposure) showed a shorter TL for a higher annual $\text{PM}_{2.5}$ exposure (Pieters et al., 2015). Similarly, chronic BC exposure has been associated with a shorter leucocyte TL in an older male study population (McCracken et al., 2010), and another study found a similar association exclusively in men (Ward-Caviness et al., 2016). In children, we also found an inverse association between chronic $\text{PM}_{2.5}$ exposure and buccal TL (Hautekiet et al., 2021). However, contrary findings emerged, such as a study showing that chronic indoor coal smoke exposure was associated with a

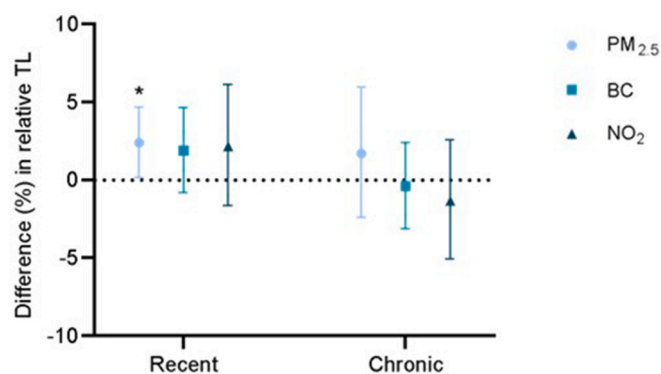


Fig. 1. Difference (%) with 95 %CI in average relative telomere length (TL) (with 95 % CI) in association with recent (week) and chronic (year) exposure to $\text{PM}_{2.5}$, BC and NO_2 . Based on multiple exposure models where the exposure windows were fitted in the same model. Estimates were adjusted for age, sex, highest educational level of the household, country of birth, region, season, smoking status, BMI and plate number as fixed covariates and for household number in the random statement. * $p < 0.05$.

Table 3

Unadjusted and adjusted analyses of the percentage change (%) in telomere length (TL) and mitochondrial DNA content (mtDNAC) in association with recent and chronic air pollutant exposure.

	TL				mtDNAC			
	Unadjusted model		Adjusted model		Unadjusted model		Adjusted model	
	% difference (95 % CI)	p-value	% difference (95 % CI)	p-value	% difference (95 % CI)	p-value	% difference (95 % CI)	p-value
PM _{2.5} recent	2.40 (0.68, 4.76)	0.009	2.40 (0.16, 4.69)	0.036	-2.39 (-4.44, -0.29)	0.025	0.29 (-2.25, 2.91)	0.82
PM _{2.5} chronic	1.97 (-0.33, 4.32)	0.094	1.70 (-2.37, 5.95)	0.42	-1.38 (-3.74, 1.01)	0.26	-3.36 (-7.87, 1.36)	0.16
BC recent	-0.37 (-2.47, 1.77)	0.73	1.90 (-0.80, 4.68)	0.17	-0.87 (-3.09, 1.4)	0.45	0.52 (-2.56, 3.70)	0.74
BC chronic	0.95 (-1.1, 3.03)	0.36	-0.38 (-3.12, 2.43)	0.79	-2.51 (-4.59, -0.38)	0.021	-3.11 (-6.19, 0.07)	0.055
NO ₂ recent	-0.23 (-2.33, 1.93)	0.84	2.17 (-1.65, 6.14)	0.27	-1.72 (-3.93, 0.52)	0.13	1.60 (-2.81, 6.19)	0.48
NO ₂ chronic	0.60 (-1.46, 2.69)	0.57	-1.31 (-5.05, 2.57)	0.50	-2.22 (-4.32, -0.06)	0.044	-4.02 (-8.22, 0.36)	0.072

Generalised linear mixed models were used to assess the association between the air pollutant and telomere length (TL) or mitochondrial DNA content (mtDNAC) with both exposure windows in the same model, adjusted for household number in the random statement. The adjusted model also included age, sex, highest educational level of the household, country of birth, region, season, smoking status, BMI, and plate number as fixed covariates. Results are presented as the % difference (95 % confidence intervals) in TL or mtDNAC for an IQR increment in the air pollutant. n = 756.

higher leucocyte TL (Xue et al., 2020). Additionally, a large-scale contemporary cohort with over 400,000 biobanked samples found no significant association between leucocyte TL and several air pollutants (Bountziouka et al., 2023). Variations in exposure assessment methods (questionnaire vs modelling) and the spatial-temporal resolution of the modelled exposures may explain these discrepancies.

In our study, higher recent air pollution exposure was associated with a longer TL, but existing literature with recent exposures is inconclusive. No association with TL was found for PM_{2.5} or NO₂ among diabetes patients (Xia et al., 2015) or older adults (Pieters et al., 2015). Conversely, higher concentrations of particulate matter were associated with a longer leucocyte TL in truck drivers and office workers (predominantly men) (Hou et al., 2012) and male steelworkers (Dioni et al., 2011). On the other hand, higher recent NO₂ concentrations were associated with a lower leucocyte TL in women (Everson et al., 2020). For women using biomass stoves, BC and PM_{2.5} exposure was associated with a shorter buccal TL (Li et al., 2019). In our previous study in children, we found shorter buccal TL with recent exposures to BC and NO₂ (Hautekiet et al., 2021), in contrast to the present study in adults.

As part of the telomere-mitochondrial ageing hypothesis, air pollution and mitochondrial-induced ROS may promote telomere attrition (Martens and Nawrot, 2016). Two systematic reviews provide evidence for an inverse association between air pollution and TL (Zhao et al., 2018), although they also suggest potentially different associations for recent versus chronic exposures (Miri et al., 2019). Similar to mtDNAC, the effects of air pollution on TL may be more pronounced in the long term, with different pathways critical for recent exposures. Several mechanisms have been proposed to explain the positive association between air pollution and TL. The first hypothesis includes inflammation, which may cause an increase in inflammatory cells (Chen and Schwartz, 2008; Salvi et al., 1999). During acute inflammation, less mature leucocytes are present in the blood (Tan et al., 2000). Consequently, an acute expansion of cells necessary to build up an inflammatory response may be associated with a longer leucocyte TL (Weng et al., 1997). A second hypothesis is the activation of telomerase. TL in leucocytes is dynamic and depends on the balance of processes that shorten telomeres and telomerase activities that elongate telomeres (Smogorzewska and de Lange, 2004; Teixeira et al., 2004). Acute inflammatory processes have been associated with elevated telomerase activity in B-cells (Weng et al., 1997). However, Dioni et al. (2011) did not replicate these findings when examining the expression of human telomerase reverse transcriptase (hTERT), a key component of telomerase (Dioni et al., 2011). A third hypothesis is the function of telomere-associated proteins. These proteins regulate telomere function by affecting telomere length or altering capping function (Blackburn, 2001). However, they are very complex, and their underlying mechanisms are not yet fully understood (Blackburn, 2001; Hodes et al., 2002).

Taken together, our findings suggest distinct associations between

recent and chronic exposures, potentially due to different mechanisms operating at different exposure windows. The existing literature on the associations between air pollution and ageing biomarkers remains limited and inconsistent. Variations in pollutant concentrations, study populations, tissues examined, and molecular techniques applied may contribute to these inconclusive results. Further research is necessary to confirm our findings and develop standardised strategies to facilitate future comparisons.

4.3. Strengths and limitations

Firstly, air pollution is modelled at the participant's residence. Although an increasing number of environmental health studies use modelled residential air pollution, it might not fully represent personal exposure. The risk estimates may be affected by exposure misclassification, potentially leading to either underestimation or overestimation, depending on participants' mobility or behaviour patterns (Shekarrizfard et al., 2016). Our previous research has demonstrated that internal black carbon load reflects residential air pollution exposure (Bongaerts et al., 2022; Saenen et al., 2017). Secondly, the cross-sectional design limited our ability to infer causality. Third, time-varying factors such as temperature and relative humidity can influence the association between air pollution and ageing markers, particularly in recent exposure windows. However, we lacked access to this data with sufficient spatial and temporal resolution for these factors. Fourth, we adjusted our main model for several covariates associated with TL and/or mtDNAC in the literature. Although age and sex showed clear inverse associations with both TL and mtDNAC, no association was found with smoking status, which was confirmed by urinary cotinine. This could be explained by the relatively small sample size of this study. Furthermore, we cannot exclude that our findings could be influenced by an unknown factor associated with TL or mtDNAC. For example, we did not have data available on blood cell counts, which in previous research has been associated with mtDNAC (Knez et al., 2016) and short-term variation in telomere length (Rehkopf et al., 2016; Weng et al., 1997). However, a key strength of this study is its assessment of two biomarkers of biological ageing across multiple exposures and time windows. Finally, we adjusted our main model for the highest educational level of the household rather than the participants' educational level. The HuBiHIS study aimed to enhance existing datasets for human biomonitoring, and for this study, we relied on available data from the BHIS study. Each person's socio-economic status (SES) was estimated by the highest level of education in the household. Assessing SES at the household level was more robust, considering that part of the population are students. Overall, different household members share material resources, lifestyle behaviours, opportunities, and networks.

5. Conclusion

In this cross-sectional study, we showed for air pollution concentrations below the annual EU standards (25 $\mu\text{g}/\text{m}^3$ and 40 $\mu\text{g}/\text{m}^3$ for $\text{PM}_{2.5}$ and NO_2 , respectively) that chronic ambient residential exposure showed a trend with lower mtDNA. In contrast, recent air pollution exposure was associated with a longer leucocyte TL. These findings indicate the possibility of different underlying mechanisms for recent and chronic exposures. Future studies are warranted to clarify these mechanisms and confirm our results.

CRedit authorship contribution statement

Pauline Hautekiet: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Tim S. Nawrot:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Dries S. Martens:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Esmée M. Bijns:** Writing – review & editing, Formal analysis. **Sigrid C.J. De Keersmaecker:** Writing – review & editing, Resources. **Johan Van der Heyden:** Writing – review & editing, Project administration, Investigation, Data curation. **Eva M. De Clercq:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Nelly D. Saenen:** Writing – original draft, Supervision, Project administration, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2025.121525>.

Data availability

Data will be made available on request.

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